

OpenSeismo: Open-Source Decentralized LoRa & 4G-Based Low-Latency Edge-AI Early Earthquake Detection and Monitoring System

Bill of Materials (BOM)

This document lists the primary hardware components used in the OpenSeismo project. Consumables, cases, and prototype boards have been excluded to reflect the absolute minimum viable hardware cost.

S.No	Component	Qty	Unit Price (₹)	Total Price (₹)	Purpose / Notes
1	Raspberry Pi 4 Model B (4GB RAM)	1	4,000	4,000	Master Node / Edge AI Processing (Used/Refurbished Pricing)
2	Quectel EC200U 4G/GSM & GPS Module	1	2,689	2,689	Cellular Fallback & Location Tracking
3	SX1278 LoRa Module (Ra-02 433MHz)	2	349	698	Long Range Mesh Communication (Pi & ESP)
4	ESP8266 NodeMCU Board	1	244	244	Remote Community Node Controller
5	MPU-6050 3-Axis Accelerometer	1	169	169	Primary Seismic Detection (STA/LTA)
6	AHT20 + BMP280 Module	1	129	129	Environmental telemetry (Temp/Hum/Pressure)
7	0.96" I2C OLED Display (Blue)	1	181	181	Live status display for ESP8266 Slave Node
8	433MHz LoRa Antenna + IPEX Connectors	2	136	272	Antennas for LoRa Modules
9	Passive Buzzer Module	2	25	50	Earthquake Audible Alarms (Master & Slave)

Estimated Minimum Viable Cost: ₹ 8,432